

Bleached+FluCoral

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Abstract

Bleached+FluCoral is a project that integrates eco-acoustic research and artistic practice to raise awareness about the deterioration of coral reefs. The project aims to highlight the connection between human action and environmental fragility, encouraging people to recognize how any interaction, even the smallest, can be a source of disturbance and contribute to the damage to coral reefs.

1 FluCoral Project

1.1 Comparative Sound Analysis

By training a FluCoMa’s neural network on sound recordings of both healthy and impacted coral reefs, we can identify distinct acoustic signatures that estimate the overall health status of the coral ecosystem. This comparative analysis is extremely useful for adaptive management strategies, allowing to give an estimate of the health status of the reef without the need for a professional diver.

1.2 Soundscape of Healthy and Impacted Reefs

Spectral Patterns: Healthy reefs display a broad spectrum of activity: high-frequency energy from shrimp (>2kHz) and low-frequency fish calls (100–1000Hz). In degraded reefs, the spectrogram is largely empty in the fish-call bands – often only the occasional shrimp snap or environmental noise remains. On a healthy reef the 2–5kHz range lights up with shrimp snaps, while fish calls fill 150–600Hz; remove the fish and the low bands go quiet.

1.3 Biological Sounds – Healthy vs. Degraded

Snapping Shrimp (“Crackle/Frying”): these tiny crustaceans produce loud broadband snaps. On a healthy reef the constant cacophony of hundreds of shrimp can dominate the soundscape (to our ears, often likened to frying bacon). On a degraded reef, researchers often find fewer shrimp snaps – the din is “substantially decreased” as the reef loses structure and hiding places. Fish Vocalizations (Low–Mid Frequencies): Many reef fishes (e.g. damselfish, grunts,

goatfish) make calls or bangs mostly below circa 1 kHz. Healthy reefs feature multiple vocalizing species; one study identified approximately 10 distinct fish sounds (e.g. grunts, chirps, knocks) on a thriving reef. These appear as mid-to low-frequency bands on spectrograms. On degraded reefs, such species are often absent or rare, so their characteristic choruses disappear Lamont et al. (2021) found phonetic richness (number of biotic sound types) to be significantly higher on healthy than on degraded reefs, reflecting many more fish calls and other sounds on the healthy sites.

1.4 Data Collection

To ensure optimal training of the neural network, one should collect soundscape data covering all possible variables (health status, season, time of day, human activity, depth, ecc.) What we did instead was using audio recordings of coral reefs situated in Indonesia, at different depths, and use those to train the FluCoMa neural network To test the quality of the training and the sound classification capabilities , we used audio recordings from different coral reefs, at different depths, situated in the Philippines. Nonetheless, we managed to properly train the neural network, obtaining an optimal fit, and, as shown in tests, succeeding in identifying the health status of the tested reefs. The FluCoMa neural network analyzes recorded soundscapes, comparing them with audio samples from both healthy and impacted reefs to assess health status.

1.5 Future Directions and Research Opportunities

Using the FluCoMa neural network is, compared to other AI powered methods, a cost and resource effective way to give a reliable estimate of the health status of a coral reef. This can allow to cut expenses in the conservation and restoration projects, without having to rely on more specialized technology or data center. Also, this method does not constantly rely on divers, significantly reducing the cost of conservation and restoration projects, as divers can then be employed only when there is a reasonable doubt on the status of the reef, instead of needing them to give the first estimate of the reef's health status. Also, using this cheap and efficient monitoring method can give a new approach to the conservation effort, as it opens the possibility to check the status of the reef to basically anyone with a computer and an hydrophone, allowing for a more decentralized conservation effort, helping involve local populations in said efforts, and reducing the reliance on big companies and big funding in these efforts.

2 Installation Concept

“B L E A C H E D.” is the multimedia installation of the project, designed to visualize and sound the negative impact that human interaction has on the health of coral reefs. Bleaching is the term used to describe the degradation of coral, which, from teeming with life, vibrant in color and sound, declines

into a faded, dull, desolate landscape. This process, inextricably linked to the presence and activity of humans, occurs silently, almost imperceptibly until it is complete, when all that remains are the grave consequences of humanity’s lack of care and respect. To raise awareness on the subject, we chose to take the opposite path, making the death process of all the organisms that make up and maintain the biologically active ecosystem of a coral reef almost unbearably deafening, anxiety-inducing, and disturbing. At the same time, we provided a clear visual representation of the result of this process: its degradation, the dissolution of a complex network of animal and plant life forms.

3 Bibliography

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